

$$T(n) - 2T(n-1) = n2^n$$

Let

$$x(n) = T(n)u(n)$$

$$\Rightarrow x(0) = 1, \quad x(1) = T(1) \dots \text{and so on}$$

$$x(n) = 0 \quad \text{for } n < 0$$

$$\Rightarrow x(n) - 2x(n-1) = n2^n u(n) + \delta(n)$$

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$